

30.08.2026

Q2)

$$(*) (7-14)_{10} = (?)_2$$

$$(7)_{10} = (0000111)_2$$

$$(14)_{10} = (00001110)_2$$

$$1's \text{ com} = 11110001$$

+1

$$2's \text{ com} = 11110010$$

$$\therefore (7)_{10} = 00000111$$

$$(-14)_{10} = 11110010$$

$$11111001$$

Again,

$$11111001$$

$$1's \text{ com} = 00000110$$

+1

$$2's \text{ com} = 00000111$$

$$= (111)_2$$

$$= (7)_{10}$$

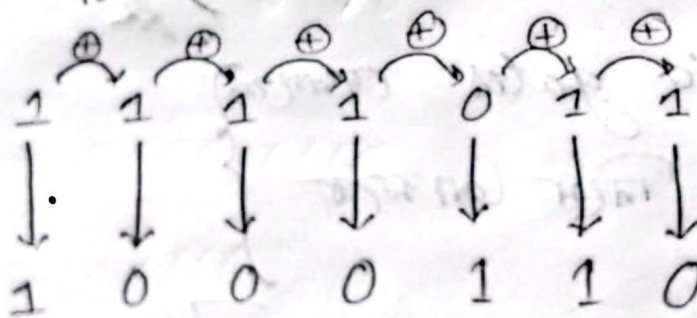
$$\therefore (7-14)_{10} = (1111001)_2$$

GRAY Code:

(*) Truth Table of X-OR:

Input		Output
A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

4) $(1234567)_{10} = (1111011)_2$

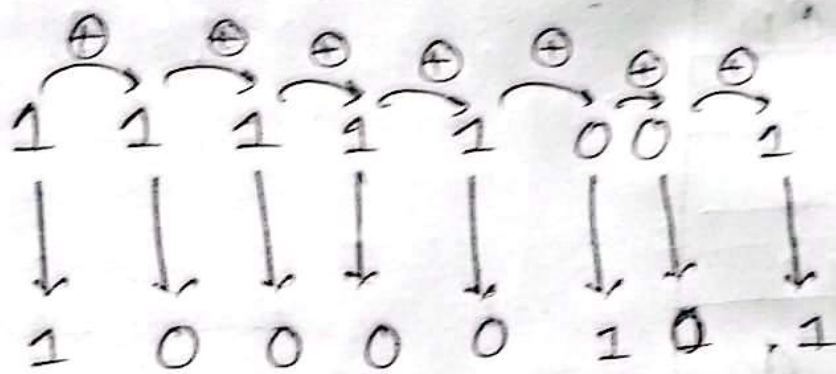


← GRAY
CODE
CONVERSION

$= (1000110)_{\text{GRAY CODE}}$

(*) $(49)_{10} = \text{101} (?)_{\text{GRAY CODE}}$

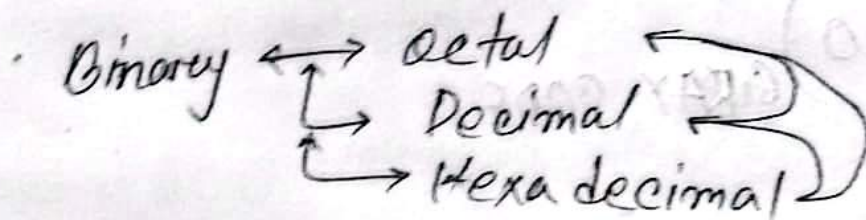
$\Rightarrow (249)_{10} = (11111001)_2$



$\therefore (249)_{10} = (10000101)_{\text{GRAY CODE}}$

{ Next day logic gates theory }
 { preview with carrier }
 (A)

How: Converts revise:



(harsh 3/2)